«part def»

Specified Nanosatellite EPS Engineering Framework :> 'Measured Nanosatellite EPS Engineering Framework'

doc

Complete OOSEM black-box definition. Inherited features provide six boundary service functions, six typed ports, eleven measures of performance, and the operational control state machine. The F-01 service includes external-artifact registration as a cross-phase configuration-management responsibility. The features below add required stores and externally declared execution resource characteristics. Internal components remain unspecified.

attributes

^<MOP-01> valid trace coverage : Percent
^<MOP-02> reference analysis reproduction success : Percent
^<MOP-03> open critical discrepancy count : Count
^<MOP-04> completed contrasting case count : Count
^<MOP-05> undeclared adaptation count : Count
^<MOP-06> correct change turnaround time : Hours
^<MOP-07> case adaptation effort : PersonHours
^<MOP-08> independent reproduction effort : PersonHours
^<MOP-09> framework adoption effort : PersonHours
^<MOP-10> direct framework cost : CAD
^<MOP-11> consistency assurance effectiveness : Percent
declared minimum execution environment : String
declared runtime dependencies : String

«ref part»

^systemModel : 'Versioned System Model'

ports

^frameworkControl : FrameworkBoundaryInterfaces::'Model Control Port'

«ref part»

^evidenceRepository : 'Evidence Repository'

ports

^frameworkCustody : FrameworkBoundaryInterfaces::'Evidence Custody Port'

«ref part»

^analysisTools : 'Analysis Tools'

ports

^frameworkExecution : FrameworkBoundaryInterfaces::'Analysis Execution Port'

«perform action»

^maintainEvidenceChain : FrameworkBlackBoxFunctions::'Maintain Evidence Chain'

items

^request : 'Evidence Chain Request'
^modelRecords : 'Controlled Model Records'
^evidenceRecords : 'Retained Evidence Records'
^registrations : 'Artifact Registration Set'
^scope : 'Evidence Chain Scope'
^assessment : 'Evidence Chain Assessment'

actions

^registerExternalArtifacts : 'Register External Engineering Artifacts'
^scopeEvidenceChain : 'Scope Evidence Chain'
^validateEvidenceChain : 'Validate Evidence Chain'
^issueAssessment : 'Issue Evidence Chain Assessment'

«#requiredStore ref item»

requiredRecords : 'Controlled Framework Record Set'

items

^modelRecords : 'Controlled Model Records' [0..*]
^evidenceRecords : 'Retained Evidence Records' [0..*]
^caseConfigurations : 'Controlled Case Configuration' [0..*]
^evaluationManifest : 'Framework Evaluation Manifest'
^artifactRegistrations : 'Artifact Registration Record' [0..*]
^chainAssessments : 'Evidence Chain Assessment' [0..*]
^reproductionVerdicts : 'Reproduction Verdict' [0..*]
^changePackages : 'Change Evidence Package' [0..*]
^comparisonPackages : 'Candidate Comparison Package' [0..*]
^reviewPackages : 'Review Package' [0..*]
^reviewStatuses : 'Review Status' [0..*]
^transferVerifications : FrameworkBoundaryInterfaces::'Transfer Verification Record' [0..*]

«perform action»

^configureFrameworkForCase : FrameworkBlackBoxFunctions::'Configure Framework for EPS Case'

items

^request : 'Case Configuration Request'
^commonCore : 'Common Core Package'
^adaptationDecision : 'Adaptation Decision'
^draft : 'Case Configuration Draft'
^configuration : 'Controlled Case Configuration'
^status : 'Configuration Status'

actions

^captureConfiguration : 'Capture Case Configuration'
^validateConfiguration : 'Validate Case Configuration'
^issueStatus : 'Issue Configuration Status'

«perform action»

^reproduceDesignatedAnalysis : FrameworkBlackBoxFunctions::'Reproduce Designated Analysis'

items

^request : 'Reproduction Request'
^modelRecords : 'Controlled Model Records'
^evidenceRecords : 'Retained Evidence Records'
^returnedResult : 'Analysis Result Package'
^executionRequest : 'Analysis Execution Request'
^result : 'Analysis Result Package'
^verdict : 'Reproduction Verdict'

actions

^preparePackage : 'Prepare Reproduction Package'
^coordinateAnalysis : 'Coordinate Controlled Analysis'
^evaluateResult : 'Evaluate Reproduced Result'
^issueVerdict : 'Issue Reproduction Verdict'

«perform action»

^supportEarlyDesignReview : FrameworkBlackBoxFunctions::'Support Early Design Review'

items

^request : 'Design Review Request'
^modelRecords : 'Controlled Model Records'
^evidenceRecords : 'Retained Evidence Records'
^findings : 'Review Findings'
^disposition : 'Finding Disposition'
^outcome : 'Review Outcome'
^reviewPackage : 'Review Package'
^issuedFindings : 'Review Findings'
^reproduction : 'Reproduction Verdict'
^status : 'Review Status'

actions

^assemblePackage : 'Assemble Review Package'
^coordinateReproduction : 'Coordinate Selected Reproduction'
^manageFindings : 'Manage Review Findings'
^issueStatus : 'Issue Review Status'

«perform action»

^assessControlledDesignChange : FrameworkBlackBoxFunctions::'Assess Controlled Design Change'

items

^request : 'Design Change Request'
^modelRecords : 'Controlled Model Records'
^evidenceRecords : 'Retained Evidence Records'
^revisions : 'Change Revision Package'
^returnedResult : 'Analysis Result Package'
^executionRequest : 'Analysis Execution Request'
^impact : 'Change Impact Assessment'
^evidencePackage : 'Change Evidence Package'

actions

^validateRequest : 'Validate Change Request'
^analyzeImpact : 'Analyze Change Impact'
^coordinateAnalysis : 'Coordinate Controlled Analysis'
^validateEvidence : 'Validate Regenerated Evidence'
^assembleEvidence : 'Assemble Change Evidence Package'

«perform action»

^compareCandidateArchitectures : FrameworkBlackBoxFunctions::'Compare Candidate EPS Architectures'

items

^request : 'Candidate Comparison Request'
^protocol : 'Comparison Protocol'
^candidates : 'Candidate Model Package'
^returnedResults : 'Analysis Result Package'
^sensitivityAndUncertainty : 'Sensitivity and Uncertainty Assessment'
^executionRequest : 'Analysis Execution Request'
^result : 'Analysis Result Package'
^comparisonPackage : 'Candidate Comparison Package'

actions

^validateBasis : 'Validate Comparison Basis'
^coordinateAnalysis : 'Coordinate Controlled Analysis'
^determineFeasibility : 'Determine Candidate Feasibility'
^compareCandidates : 'Compare Candidate Architectures'
^issueComparison : 'Issue Candidate Comparison'

«exhibit state»

^operationalControl : FrameworkOperationalState::'Framework Operational Control'

states

^<ST-01> Unconfigured
^<ST-02> Configuring
^<ST-03> Controlled Baseline Ready
^<ST-04> Assessing Evidence
^<ST-05> Reproducing Analysis
^<ST-06> Assessing Change
^<ST-07> Comparing Candidates
^<ST-08> Under Review
^<ST-09> Discrepancy Open
^<ST-10> Recovering

state transition

^initializeControl
^<T-01> beginConfiguration
^<T-02> configurationAccepted
^<T-03> configurationDeficient
^<T-04> reconfigureContext
^<T-05> assessEvidenceRequest
^<T-06> reproduceAnalysisRequest
^<T-07> assessChangeRequest
^<T-08> compareCandidatesRequest
^<T-09> reviewRequest
^<T-10> evidenceAccepted
^<T-11> evidenceDeficient
^<T-12> reproductionAccepted
^<T-13> reproductionDeficient
^<T-14> changeAccepted
^<T-15> changeDeficient
^<T-16> comparisonAccepted
^<T-17> comparisonDeficient
^<T-18> reviewAccepted
^<T-19> reviewDeficient
^<T-20> beginControlledRecovery
^<T-21> retryConfiguration
^<T-22> retryReproduction
^<T-23> retryChange
^<T-24> retryComparison
^<T-25> retryReview
^<T-26> recoveryAccepted
^<T-27> recoveryDeficient

^missionApplication : FrameworkBoundaryInterfaces::'Mission Application Port'
^researchGovernance : FrameworkBoundaryInterfaces::'Research Governance Port'
^independentReview : FrameworkBoundaryInterfaces::'Independent Review Port'
^modelControl : FrameworkBoundaryInterfaces::'Model Control Port'
^analysisExecution : FrameworkBoundaryInterfaces::'Analysis Execution Port'
^evidenceCustody : FrameworkBoundaryInterfaces::'Evidence Custody Port'